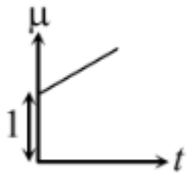
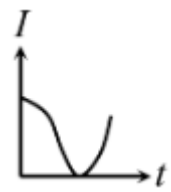
Quizrr Chapter-Wise Test for JEE Advanced - 2025

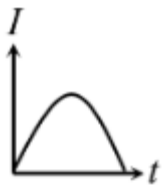
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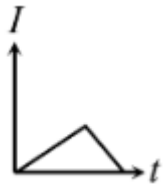
Q.1

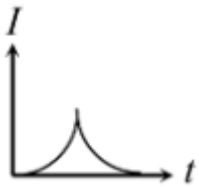
In YDSE, one of the slits is covered by a thin sheet of variable refractive index, whose variation with time is represented by graph as shown. The variation of intensity (I) at a point on the screen with time, where first minima is obtained at $t = 0$ will be best represented by



(A) 

(B) 

(C) 

(D) 

Q.2

In Young's double slit experiment, double slit of separation 0.1 cm is illuminated by white light. A coloured interference pattern is formed on a screen 100 cm away. If a pin hole is located on this screen at a distance of 2 mm from the central fringe, the wavelength in the visible spectrum which will be absent in the light transmitted through the pin-hole are

(A) $5714\mathring{A}$ and $4444\mathring{A}$

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(B) $6000\mathring{A}$ and $5000\mathring{A}$

(C) $5500\mathring{A}$ and $4500\mathring{A}$

(D) $5200\mathring{A}$ and $4200\mathring{A}$

Q.3

Young's double slit experiment is first performed in air and then by immersing the whole setup in a liquid. The 10th bright fringe when in liquid is formed at the point where 8th dark fringe is located when in air. The refractive index of the liquid is :

(A) 1.25

(B) 1.33

(C) 1.40

(D) 1.20

Q.4

In a Young's experiment, two coherent sources are placed $0.90\ \text{mm}$ apart and the fringes are observed one metre away. It produces the second dark fringe at a distance of $1\ \text{mm}$ from the central fringe, the wavelength of monochromatic light used would be.

(A) $60 \times 10^{-4}\ \text{cm}$

(B) $10 \times 10^{-4}\ \text{cm}$

(C) $10 \times 10^{-5}\ \text{cm}$

(D) $6 \times 10^{-5}\ \text{cm}$

Q.5

The young's double slits experiment is performed with blue and with green light of wavelength $4360\mathring{A}$ and $5460\mathring{A}$ respectively. It x is the distance of the 4th maxima from the central one, then

(A) $x_{\text{blue}} = x_{\text{green}}$

(B) $x_{\text{blue}} > x_{\text{green}}$

(C) $x_{\text{blue}} < x_{\text{green}}$

(D) None of the above

Q.6

A beam of light consisting of two wavelength $6500\mathring{A}$ & $5200\mathring{A}$ is used to obtain interference fringes in a young's double slit experiment. The distance between the slits is $2.0\ \text{mm}$ and the distance between the plane of the slits and the screen is $120\ \text{cm}$. What is the least distance from the central maximum where the bright fringes due to both the wave length coincide ?

(A) $0.156\ \text{cm}$

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(B) 0.152 cm

(C) 0.17

(D) 0.16 cm.

Q.7

In an experiment similar to young's experiment, interference is observed using waves associated with electrons. The electrons are being produced in an electron gun. In order to increase the fringe width.

(A) electron gun voltage be increased.

(B) electron gun voltage be decreased.

(C) the slit be moved away.

(D) the screen be moved closer to interfering slits.

Q.8

Interference fringes were produced in young's double slit experiment using light of wave length $5000\mathring{A}$. When a film of material 2.5×10^{-3} cm thick was placed over one of the slits, the fringe pattern shifted by a distance equal to 20 fringe width. The refractive index of the material of the film is-

(A) 1.25

(B) 1.33

(C) 1.4

(D) 1.5

Q.9

If the first minima in a Young's slit experiment occurs directly in front of one of the slits, (distance between slit and screen $\mathrm{D} = 12$ cm and distance between slits $\mathrm{d} = 5$ cm.) then the wavelength of the radiation used can be -

(A) 2 cm.

(B) 4 cm.

(C) $2/3$ cm.

(D) $4/3$ cm.

Q.10

Choose the correct options -

(A) A point object is kept in front of a plane mirror. The plane mirror is performing SHM of amplitude 2 cm. The plane mirror moves along the x-axis and y-axis is normal to the mirror. The amplitude of the mirror is such that the object is always infront of the mirror. The amplitude of SHM of the image is 4 cm.

(B) In Young's double slit experiment 7 maxima can be obtained on screen including central maxima if $\mathrm{d} = 7000\mathring{A}$ and $\lambda = 2000\mathring{A}$.

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(C) A convex lens of focal length $30\ \text{cm}$ and a concave lens of focal length $10\ \text{cm}$ are placed at some distance. If a parallel beam of light incident on convex lens emerges as parallel beam from convex lens than the distance between two lenses is $20\ \text{cm}$.

(D) In Young's double slit experiment 5 maxima can be obtained on screen including central maxima if $\mathrm{d} = 7000\mathring{A}$ and $\lambda = 2000\mathring{A}$.

Q.11

Two coherent point light sources are placed at $x = -d/2$ and $x = d/2$. These sources are surrounded by a spherical screen having equation $x^3 + y^2 + z^2 = R^2 (R \gg d)$. The sources emit monochromatic light of wavelength $\lambda = \mathrm{d}/3$

(A) The number of maxima produced on the screen are 7

(B) The shape and length of second order maxima is circle, $\frac{2\pi \mathrm{R}\sqrt{5}}{3}$

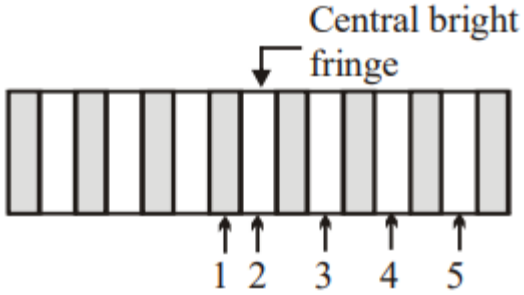
(C) If radius of screen is doubled the intensity of maxima will become one fourth

(D) All maxima are equally spaced along x-axis

Q.12

Passage

The figure shows the interference pattern obtained in a double-slit experiment using light of wavelength $600\ \text{nm}.1, 2, 3, 4$ and 5 are marked on five fringes.



Question

The third order bright fringe is -

(A) 2

(B) 3

(C) 4

(D) 5

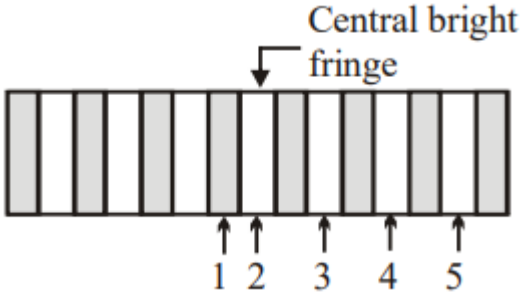
Q.13

Passage

The figure shows the interference pattern obtained in a double-slit experiment using light of wavelength

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600 nm.1, 2, 3, 4 and 5 are marked on five fringes.



Question

Which fringe results from a phase difference of 4π between the light waves incidenting from two slits -

(A) 2

(B) 3

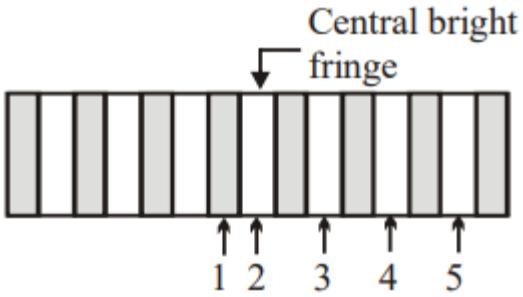
(C) 4

(D) 5

Q.14

Passage

The figure shows the interference pattern obtained in a double-slit experiment using light of wavelength 600 nm.1, 2, 3, 4 and 5 are marked on five fringes.



Question

Let ΔX_A and ΔX_C represent path differences between waves interfering at 1 and 3 respectively then $(|\Delta X_C| - |\Delta X_A|)$ is equal to -

(A) 0

(B) 300 nm

(C) 600 nm

(D) 900 nm

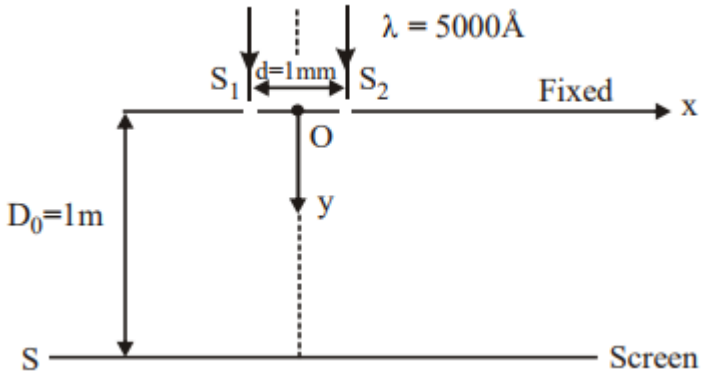
Q.15

Passage

In the figure shown light of wavelength $\lambda = 5000\mathring{A}$ is incident on the slits (in a horizontal fixed plane) S_1 and S_2 separated by distance $d = 1$ mm. A horizontal screen S is released from rest from initial distance $D_0 = 1$ m

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from the plane of the slits. Taking origin at O and positive x and y axis as shown, (Use $g = 10\ \mathrm{m/s^2}$)



Question

Velocity of central maxima in vector form at $\mathrm{t} = 2\mathrm{sec}$. is $-$

(A) $20\hat{\mathrm{j}}\mathrm{m/s}$

(B) $5\hat{\mathrm{j}}\mathrm{m/s}$

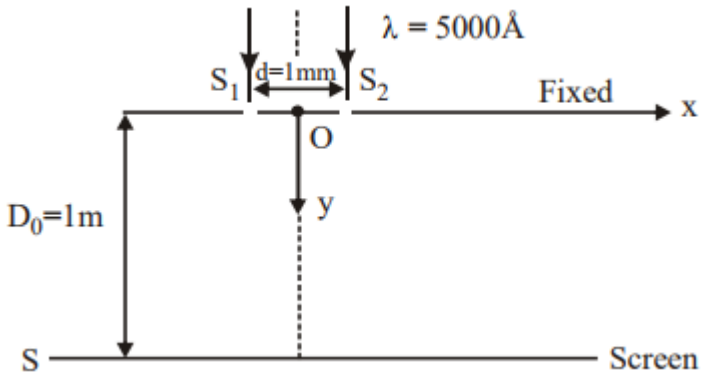
(C) $10\hat{\mathrm{j}}\mathrm{m/s}$

(D) $25\hat{\mathrm{j}}\mathrm{m/s}$

Q.16

Passage

In the figure shown light of wavelength $\lambda = 5000\mathring{A}$ is incident on the slits (in a horizontal fixed plane) S_1 and S_2 separated by distance $d = 1\ \mathrm{mm}$. A horizontal screen S is released from rest from initial distance $\mathrm{D_0} = 1\ \mathrm{m}$ from the plane of the slits. Taking origin at O and positive x and y axis as shown, (Use $g = 10\ \mathrm{m/s^2}$)



Question

Acceleration of central maxima in vector form at $\mathrm{t} = 2\mathrm{sec}$. is-

(A) $5\hat{j}\ \mathrm{m/s^2}$

(B) $15\hat{\mathrm{j}}\mathrm{m/s^2}$

(C) $10\hat{\mathrm{j}}\mathrm{m/s^2}$

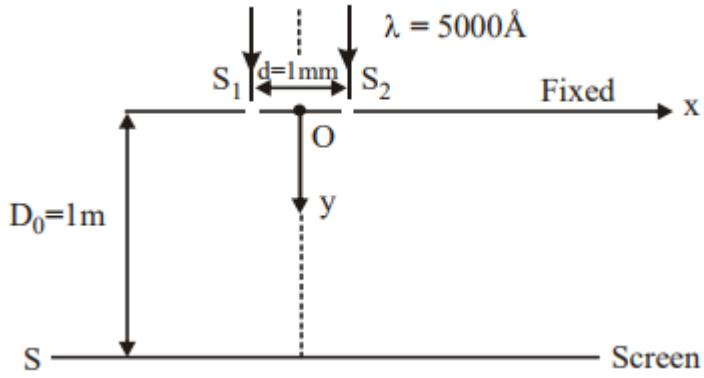
(D) $20\hat{\mathrm{j}}\mathrm{m/s^2}$

Q.17

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Passage

In the figure shown light of wavelength $\lambda = 5000\mathring{A}$ is incident on the slits (in a horizontal fixed plane) S_1 and S_2 separated by distance $d = 1\ \text{mm}$. A horizontal screen S is released from rest from initial distance $\mathrm{D_0} = 1\ \mathrm{m}$ from the plane of the slits. Taking origin at O and positive x and y axis as shown, (Use $g = 10\ \mathrm{m/s^2}$)



Question

Relative acceleration of second maxima with respect to first minima, on the same side of central maxima is -

(A) $2.5 \times 10^{-3}\ \mathrm{m/s^2}\hat{\mathrm{i}}$

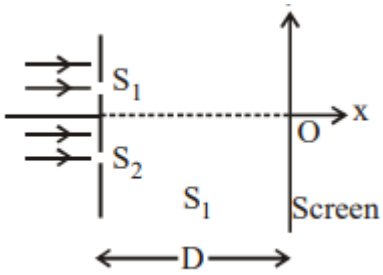
(B) $7.5 \times 10^{-3}\ \mathrm{m/s^2}\hat{\mathrm{j}}$

(C) $5.5 \times 10^{-3}\ \mathrm{m/s^2}\hat{\mathrm{i}}$

(D) $7.5 \times 10^{-3}\ \mathrm{m/s^2}\hat{\mathrm{i}}$

Q.18

A monochromatic parallel beam of light of wavelength λ is incident normally on the plane containing slits $\mathrm{S_1}$ and $\mathrm{S_2}$. The slits are of unequal width such that intensity only due to one slit on screen is four times that only due to other slit. The screen is placed along y axis is as shown. The distance between slits is d and that between screen and slit is D. Match the statements in column I with results in column-II.



Column I	**Column II**
(A) The distance between two points on screen having equal	(p) Dλ/3d

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intensities, such that intensity at those points is $\frac{1}{9}$th of maximum intensity.

(B) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{3}{9}$th of maximum intensity. (q) $D\lambda/d$

(C) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{5}{9}$th of maximum intensity. (r) $2D\lambda/d$

(D) The distance between two points on screen having equal intensities, such that intensity at those points is $\frac{7}{9}$th of maximum intensity. (s) $3D\lambda/d$

(A) $(A) - q, r, s (B) - p, q, r, s (C) - q, r, s (D) p, q, r, s$

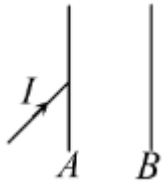
(B) $(A) - s, p (B) - p, (C) - s, (D) - q$

(C) $(A) - q, r (B) - q, r (C) - p, q, r, s (D) - p, s$

(D) $(A) - p, r, (B) - q, r, (C) - q, r, (D) - q, r$

Q.19

A narrow monochromatic beam of light of intensity I is incident on a glass plate A as shown in figure. An other identical glass plate B is kept close to A and parallel to it. Each glass plate reflects 25% of the light intensity incident on it and transmits the remaining. Find the ratio of the maximum and minimum intensities in the interference pattern formed by the two beams obtained after one reflection at each plate.



Q.20

Newton's rings are visible when a planoconvex lens is placed on a flat glass surface. For a particular lens with an index of refraction of $n = 1.50$ and a glass plate with an index of $n = 1.80$, the diameter of the third bright ring is $0.640\ \text{mm}$. If water $(n = 1.33)$ now fills the space between the lens and the glass plate, what is the new diameter of this ring? Assume the radius of curvature of the lens is much greater than the wavelength of the light.

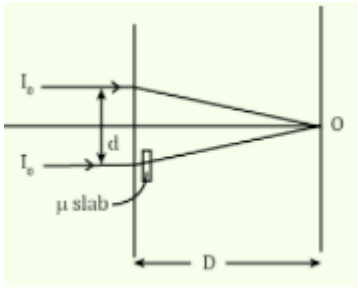
Answers & Solutions

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Q.1 Answer:

—

Solution:



The path difference due to the slab is, $(\Delta x)_0 = (\mu - 1)t \quad (t = \text{thickness})$

And its corresponding phase difference is,

$$\Delta\phi = \frac{2\pi}{\lambda}(\mu - 1)t$$

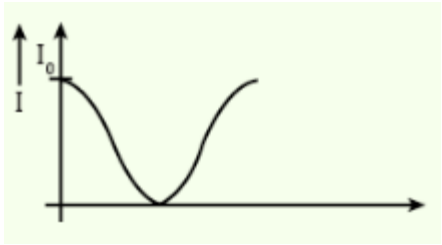
$$I_{\text{res}} = 4I_0 \cos^2 \frac{\Delta\phi}{2}$$

For $I_{\max}; \cos^2\left(\frac{\Delta\phi}{2}\right) = 1$

$$\Rightarrow \Delta\phi = 0$$

$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda}(\mu - 1)t = 0$$

$\Rightarrow \mu = 1$ **(and function will be cosine)**



Q.2 Answer:

$5714\mathring{A}$ and $4444\mathring{A}$

Solution:

Absent wavelengths correspond to interference minima

$$\therefore \quad d\sin\theta = (2n-1)\tfrac{\lambda}{2} \Rightarrow d\tfrac{y}{D} = (2n-1)\tfrac{\lambda}{2} \Rightarrow \lambda = \tfrac{2yd}{D(2n-1)} = \tfrac{40000}{2(n-1)}$$

$$= 13333\mathring{A}, 80000\mathring{A}, 5714\mathring{A}, 4444\mathring{A}, 3636\mathring{A}$$

$$\therefore \quad (\text{a})$$

Q.3 Answer:

1.33

Solution:

Fringe width $w = \frac{\lambda D}{d}$ **where symbols havo their usual meanings. Note that wavelchgth** λ **in air changes to** $\frac{\lambda}{\mu}$ **in a liquid of refractive index** μ**.**

Now, we can write $10 \times \text{w}_{\text{liq}} = 7.5 \times \text{w}_{\text{air}} \Rightarrow 10\frac{\lambda_{\text{liq}}\,\text{D}}{\text{d}} = 7.5\frac{\lambda_{\text{air}}\,\text{D}}{\text{d}} \Rightarrow 10\frac{\lambda_{\text{air}}}{\mu} = 7.5\lambda_{\text{air}}$ **.**

This then gives $\mu = \frac{10}{7.5} = 1.33$**.**

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Q.4 Answer:

6×10^{-5} cm

Solution:

$\mathrm{D} = 1$ m., $\mathrm{d} = .90$ mm $= .9 \times 10^{-3}$ m

The distance of the second dark ring from centre $= 10^{-3}$ m

$$\therefore \quad \mathrm{X_n} = (2\mathrm{n}-1)\frac{\lambda}{2}\frac{\mathrm{D}}{\mathrm{d}}$$

$$\text{For } \mathrm{n} = 2, \mathrm{X_n} = \frac{3\lambda}{2}\frac{\mathrm{D}}{\mathrm{d}} \Rightarrow \lambda = \frac{2\mathrm{Xnd}}{3\mathrm{D}} = \frac{2\times10^{-3}\times.9\times10^{-3}}{3}$$

$\lambda = 6 \times 10^{-7}$ m.

$\lambda = 6 \times 10^{-5}$ cm.

Q.5 Answer:

$x_{\text{blue}} < x_{\text{green}}$

Solution:

Distance of $\mathrm{n^{th}}$ **maximum,** $\mathrm{x} = \frac{\mathrm{n}\lambda\mathrm{D}}{\mathrm{d}}$

$$\frac{\mathrm{x_{blue}}}{\mathrm{x_{green}}} = \frac{4360}{5460} \quad \therefore \mathrm{x_{blue}} < \mathrm{x_{green}}$$

Q.6 Answer:

0.156 cm

Solution:

Suppose the $\mathrm{m^{th}}$ **bright fringe of** $6500\mathring{A}$ **coincides with the** $\mathrm{n^{th}}$ **bright fringe of** $5200\mathring{A}$

$$X_n = \frac{m\lambda_1 D}{d} = \frac{n\lambda_2 D}{d}$$

$$\Rightarrow \frac{m \times 6500 \times D}{d} = \frac{n \times 5200 \times D}{d} \Rightarrow \frac{m}{n} = \frac{5200}{6500} = \frac{4}{5}$$

$$\therefore \quad \text{Distance y is } y = \frac{m\lambda_1 D}{d} = \frac{4 \times 6500 \times 10^{10} \times 1.2}{2 \times 10^{-3}} = 0.156 \text{ cm.}$$

Q.7 Answer:

electron gun voltage be decreased.

Solution:

Fringe width $\beta = \frac{\lambda D}{d}$

According to de Broglie, wavelength $\lambda = \frac{\mathrm{h}}{\mathrm{p}} = \frac{\mathrm{h}}{\sqrt{2\mathrm{mv}}}$

As V **decreases,** λ **increases,** β **increases.**

Q.8 Answer:

1.4

Solution:

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$n = \frac{(\mu-1)tD}{d}$ **but** $\beta = \frac{\lambda D}{d} \Rightarrow \frac{D}{d} = \frac{\beta}{\lambda}$

$$n = (\mu - 1)t\beta/\lambda$$
$$20\beta = (\mu - 1)2.5 \times 10^{-3}\left\{\beta/5000 \times 10^{-8}\right\}$$
$$\mu - 1 = \frac{20 \times 5000 \times 10^{-8}}{2.5 \times 10^{-3}} = 0.4; \mu = 1.4$$

Q.9 Answer:

2 cm.

Solution:

Path difference $= \sqrt{D^2 + d^2} - D = 1$ cm.

Also, $\left[\sqrt{D^2 + d^2} - D\right] = (2n - 1)\frac{\lambda}{2} \Rightarrow \lambda = \frac{2(1)}{2n-1}$

For $n = 1, 2, 3.$

$\lambda = 2$ cm, $\frac{2}{3}$ cm, $\frac{2}{5}$ cm,

Q.10 Answer:

In Young's double slit experiment 7 maxima can be obtained on screen including central maxima if $d = 7000\mathring{A}$ and $\lambda = 2000\mathring{A}$.

Solution:

(A) Since mirror is moving perpendicular to y-axis to image will remain stationary and hence its amplitude is zero.

(B) $\Delta x = d\sin\theta = n\lambda$ **where** $n = 0, \pm 1, \pm 2 \ldots\ldots$

$\sin\theta = \frac{n\lambda}{d} = n \times \frac{2000}{7000} = \frac{2n}{7}$

$\sin\theta = 0, \pm\frac{2}{7}, \pm\frac{4}{7}, \pm\frac{6}{7}$

$\sin\theta = \frac{n\lambda}{d} = n \times \frac{20}{70}$ $\sin\theta = 0, \pm\frac{2}{7}, \pm\frac{4}{7}$ **Total** $= 7$ **maxima**

(C) $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}; \frac{1}{\infty} = \frac{1}{30} + \frac{1}{(-10)} - \frac{d}{30 \times -10}; d = 20$ cm.

Q.11 Answer:

The number of maxima produced on the screen are 7

Solution:

Order of maxima are $3, 2, 1, 0, 1, 2, 3$

Along x-axis : $\left(\frac{d}{2} + x\right) - \left(\frac{d}{2} - x\right) = n\lambda$

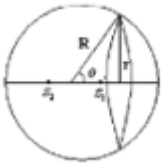
$$2\pi = n\lambda$$
$$x_1 = \frac{\lambda}{2} x_2 = \frac{2\lambda}{2} x_3 = \frac{3\lambda}{3}$$

hence $x_2 - x_1 = x_3 - x_2$ **also**

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$$d\cos\theta = n\lambda$$
$$3\lambda\cos\theta = 2\lambda$$
$$\cos\theta = \frac{2}{3}$$
$$\sin\theta = \frac{\sqrt{5}}{3}$$

$r = R\sin\theta$
$r = \frac{R\sqrt{5}}{3}$



Q.12 Answer:

5

Solution:

Order of the fringe can be counted on either side of the central maximum. For example no. 3 is first order bright fringe.

$\Delta X_C = \lambda, \Delta X_A = \frac{\lambda}{2}$
$\Delta X_C - \Delta X_A = \frac{\lambda}{2} = 300\ \text{nm}$

Q.13 Answer:

4

Solution:

Order of the fringe can be counted on either side of the central maximum. For example no. 3 is first order bright fringe.

$\Delta X_C = \lambda, \Delta X_A = \frac{\lambda}{2}$
$\Delta X_C - \Delta X_A = \frac{\lambda}{2} = 300\ \text{nm}$

Q.14 Answer:

300 nm

Solution:

Order of the fringe can be counted on either side of the central maximum. For example no. 3 is first order bright fringe.

$\Delta X_C = \lambda, \Delta X_A = \frac{\lambda}{2}$
$\Delta X_C - \Delta X_A = \frac{\lambda}{2} = 300\ \text{nm}$

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Q.15 Answer:

$20\hat{j}m/s$

Solution:

Velocity and acceleration of central maximum = velocity and acceleration of screen

[$\because$ it does not move to the left or right on the screen]

$v_{screen} = 0 + gt = 20\ m/s$

Q.16 Answer:

$10\hat{j}m/s^2$

Solution:

Velocity and acceleration of central maximum = velocity and acceleration of screen

[$\because$ it does not move to the left or right on the screen]

$v_{screen} = 0 + gt = 20\ m/s$

$\overrightarrow{v}_s = 20\hat{j}m/s$ (in vector form) and $\overrightarrow{a}_s = 10\hat{j}m/s^2$

$D = D_0 + \frac{1}{2}gt^2$[where D in distance of screen at a time t along y-axis]

Q.17 Answer:

$7.5 \times 10^{-3}\ m/s^2\hat{i}$

Solution:

$D = D_0 + \frac{1}{2}gt^2$[where D in distance of screen at a time t along y-axis]

Position vector of 2nd maximum is given by $\overrightarrow{r}_x = \pm 2\frac{\lambda D}{d}\hat{i} + D\hat{j}$

Position vector of 1st minimum is given by $\overrightarrow{r}_n = \pm\frac{1}{2}\frac{\lambda D}{d}\hat{i} + D\hat{j}$

Relative position vector of 2nd maximum with respect to 1st minimum is given by $\overrightarrow{r} = \pm\frac{3}{2}\frac{\lambda D}{d}\hat{i}$

Differentiating with respect to time, $\overrightarrow{a} = \frac{d^2\overrightarrow{r}}{dt^2} = \pm\frac{3}{2}\frac{\lambda g}{d}\hat{i} = \pm 7.5 \times 10^{-3}\ m/s^2\hat{i}$

Q.18 Answer:

$(A) - q, r, s (B) - p, q, r, s (C) - q, r, s (D) p, q, r, s$

Solution:

(A) - qrs, (B) - pqrs, (C) - qrs, (D) pqrs

Initially at a distance x from central maxima on screen is $I = I_0 + 4I_0 + 2\sqrt{I_0}\sqrt{4I_0}\cos\frac{2\pi x}{\beta}$, **where** $\beta = \frac{D\lambda}{d}$ $I_{max} = 9I_0$ **and** $I_{min} = I_0$

(A) At points where intensity is $\frac{1}{9}$ th of maximum intensity, minima is formed.

$\therefore$ **Distance between such points is** $\beta, 2\beta, 3\beta, 4\beta, \ldots\ldots$

(B) At points where intensity is $\frac{3}{9}$ th of maximum intensity, $\cos\frac{2\pi x}{\beta} = -\frac{1}{2}$ **or** $x = \frac{\beta}{3}$

(C) $\cos\frac{2\pi x}{\beta} = 0$ **or** $x = \frac{\beta}{4}$

$\therefore$ **Distance between such points is** $\frac{\beta}{2}, \beta, \beta + \frac{\beta}{2}, 2\beta, \ldots.$

(D) $\cos\frac{2\pi x}{\beta} = \frac{1}{2}$ **or** $x = \frac{\beta}{6}$

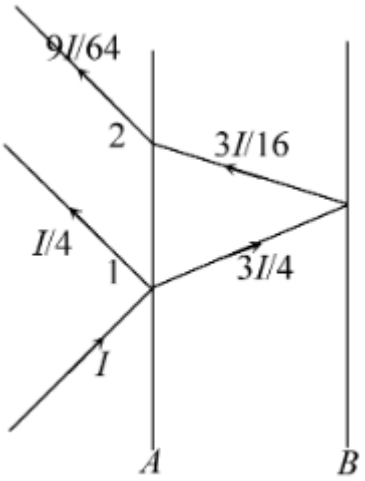
$\therefore$ **Distance between such points is** $\frac{\beta}{3}, \frac{2\beta}{3}, \beta, \beta + \frac{\beta}{3}, \beta + \frac{2\beta}{3}, 2\beta, \ldots$

Q.19 Answer:

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Solution:



$$I_1 = I \times \frac{25}{100} = \frac{I}{4}, \quad \frac{3I}{4} \times \frac{25}{100} = \frac{3I}{16}$$

$$I_2 = \frac{3I}{16} \times \frac{75}{100} = \frac{9I}{64}, I_1 = ka_1^2 \text{ and } I_2 = ka_2^2$$

$$a_1 = 4 \text{ units and } a_2 = 3 \text{ units}$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{4+3}{4-3}\right)^2 = 49$$

Q.20 Answer:

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Solution:

IDENTIFY: Consider the interference between light reflected from the top and bottom surfaces of the air film between the lens and the glass plate. Introducing a liquid between the lens and the plate just changes the wavelength from λ_0 to $\frac{\lambda_0}{n}$, where n is the refractive index of the liquid.

SET UP: For maximum intensity, with a net half-cycle phase shift due to reflections, $2t = \left(m + \frac{1}{2}\right)\lambda$, where λ is the wavelength in the film. $t = R - \sqrt{R^2 - r^2}$.

EXECUTE: $\frac{(2m+1)\lambda}{4} = R - \sqrt{R^2 - r^2} \Rightarrow \sqrt{R^2 - r^2} = R - \frac{(2m+1)\lambda}{4}$

$$\Rightarrow R^2 - r^2 = R^2 + \left[\frac{(2m+1)\lambda}{4}\right]^2 - \frac{(2m+1)\lambda R}{2} \Rightarrow r = \sqrt{\frac{(2m+1)\lambda R}{2} - \left[\frac{(2m+1)\lambda}{4}\right]^2}$$

$$\Rightarrow r = \sqrt{\frac{(2m+1)\lambda R}{2}}, \text{ for } R \gg \lambda.$$

$\lambda = \lambda_0/n$, where λ_0 is the wavelength in air. Therefore, if r_0 is the radius of the third bright ring when air is between the lens and plate, the radius with water between the lens and plate is

$$r = \frac{r_0}{\sqrt{n}} = \frac{0.640 \text{ mm}}{\sqrt{1.33}} = 0.555 \text{ mm}.$$

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Evaluate: The refractive index of the water is less than that of the glass plate, so the phase changes on reflection are the same as when air is in the space.

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